

# Heat transfer modeling for supercritical methane flowing in rocket engine cooling channels



Marco Pizzarelli <sup>a,\*</sup>, Francesco Nasuti <sup>a</sup>, Marcello Onofri <sup>a</sup>, Pietro Roncioni <sup>b</sup>,  
Raffaele Votta <sup>b</sup>, Francesco Battista <sup>b</sup>

<sup>a</sup> University of Rome "La Sapienza", Dipartimento di Ingegneria Meccanica e Aerospaziale, Via Eudossiana 18, 00184 Rome, Italy

<sup>b</sup> CIRA (Italian Aerospace Research Center), Via Maiorise, 81043 Capua, CE, Italy

## HIGHLIGHTS

- Numerical simulation of methane in supercritical pressure and subcritical to supercritical temperature.
- Fluid and structure thermal coupling.
- Asymmetrically heated channels of rectangular cross section.
- Supercritical-pressure methane may exhibit heat transfer deterioration.
- Heat transfer deterioration is mitigated increasing coolant pressure and/or surface roughness.

## ARTICLE INFO

### Article history:

Received 12 March 2014

Accepted 3 October 2014

Available online 16 October 2014

### Keywords:

Liquid rocket engine

Regenerative cooling

Supercritical methane

Heat transfer deterioration

Computational Fluid Dynamics (CFD)

Conjugate Heat Transfer (CHT)

Channel roughness

## ABSTRACT

To investigate the methane behavior inside rocket engine cooling channels, a test article has been specifically designed by the Italian Aerospace Research Center. In this study, the expected wall and coolant behavior of this test article is analyzed by a three-dimensional conjugate heat transfer model. Different coolant pressure and surface roughness levels are considered in order to understand their influence on the heat transfer capability of the cooling system. Results show that heat transfer deterioration may occur in principle when methane is operated in near-critical condition. However, the resultant wall temperature peak reduces for increasing coolant pressure and for increasing surface roughness.

© 2014 Elsevier Ltd. All rights reserved.

## 1. Introduction

The hot gas environment within a modern liquid rocket combustion chamber can be characterized by gas temperatures as large as 3600 K and heat fluxes as large as 160 MW/m<sup>2</sup>, depending on propellants, mixture ratio and chamber pressure [1]. In order to keep the temperatures of the thrust chamber walls within their allowed limits, an intense cooling effort is necessary, which, in case of regenerative cooling, is achieved by flowing one of the two propellants into suitable channels surrounding the thrust chamber. In this case, high supply pressure is required because of the inevitable losses which occur in the cooling circuit [2]. Regenerative cooling performance is especially important in case of reusable or

long-duration-expendable engines, where an effective and efficient cooling system is crucial to extend the engine life, or in expander cycle engines, where coolant heating provides the available power for turbo-machinery. In the latter case, thermal analysis of regeneratively cooled engines is essential to predict not only wall temperature but also coolant temperature and pressure at the channel exit.

The study of heat transfer to near-critical fluids, that is to supercritical pressure fluids whose temperature is close to the pseudo-critical value (i.e., the temperature at which specific heat at constant pressure has a maximum at a specified pressure), has recently captured the interest of liquid rocket engine designers because of the possible use of methane as a denser and cheaper replacement of hydrogen in launch vehicles and as a cheaper replacement of toxic storable propellants for space propulsion [3,4]. In case of liquid-oxygen/liquid-methane engines with chamber

\* Corresponding author.

E-mail address: [marco.pizzarelli@uniroma1.it](mailto:marco.pizzarelli@uniroma1.it) (M. Pizzarelli).

**Nomenclature**

|           |                                    |
|-----------|------------------------------------|
| $\dot{m}$ | mass flow rate                     |
| $b$       | channel width                      |
| $c_p$     | specific heat at constant pressure |
| $h$       | channel height                     |
| $h_c$     | heat transfer coefficient          |
| $k$       | thermal conductivity               |
| $L$       | channel length                     |
| $p$       | pressure                           |
| $q$       | heat flux                          |
| $s_w$     | internal wall thickness            |
| $T$       | temperature                        |
| $t_w$     | rib thickness                      |

$x$  axial coordinate

**Subscripts**

|     |                 |
|-----|-----------------|
| 0   | total           |
| c   | critical        |
| e   | exit            |
| i   | inlet           |
| inn | inner-interface |
| pc  | pseudo-critical |
| w   | wall            |

**Greek**

|               |                   |
|---------------|-------------------|
| $\varepsilon$ | surface roughness |
|---------------|-------------------|

pressures larger than about 50 bar, a peculiar behavior of methane as a coolant is found: it enters channels at supercritical pressure and subcritical temperature and then, under heating from hot gas, its temperature may increase up to reach and exceed the pseudo-critical value. As a consequence, because the typical pressure and temperature of methane in cooling channels are relatively close to the critical point, the coolant flow behavior is strongly affected by the large property variations that occur in the near-critical region. This behavior is often referred to as that of a “transcritical fluid” flow. In particular, variables such as specific heat at constant pressure, thermal conductivity and speed of sound exhibit a relevant peak value in the vicinity of the pseudo-critical temperature. The peak values decrease with increasing pressure; consequently, for sufficiently large pressure levels the influence of the near-critical region on the flow behavior vanishes.

To better understand the fluid-dynamic phenomena that occur in case of transcritical fluids, it is interesting to compare the thermodynamic behavior of a transcritical fluid with that of fluid boiling at subcritical pressure. In the latter case, under wall heating, the fluid phase changes from liquid to vapor, with an abrupt variation of its thermodynamic properties. More specifically, in case of low heat flux nucleate boiling occurs and wall heat transfer increases, whereas in case of high heat flux a gaseous film insulates the hot wall (film boiling), leading to a progressively increasing wall temperature. In case of supercritical pressure the phase-change does not occur. However, if pressure is rather close to the critical point, in case of wall heating the fluid passes from a liquid-like state to a gas-like state: a pseudo phase-change takes place. As a result, heat transfer deterioration may occur in case of low mass flow rate, high heat flux, and fluid and wall temperature respectively lower and higher than the pseudo-critical value [5,6].

Although a number of studies have been conducted on supercritical hydrogen flow in cooling channels to investigate the thermal behavior of LOX/LH<sub>2</sub> rocket engines, only few experimental investigations on supercritical-methane flow inside cooling channels can be found in the literature [7–9]. Moreover, apart the very few data presented in Ref. [8], these studies do not refer to rectangular-cross-section cooling channels, which are of great interest in rocket engine applications. As for numerical computations, supercritical-methane flow in both heated circular-cross-section tubes and rectangular-cross-section cooling channels have been studied by means of two-dimensional (e.g., [10,11]) and three-dimensional (e.g., [12–14]) CFD models.

The Italian interest on methane as coolant for liquid rocket engine cooling channels has led to the HYPROB program, which is carried out by CIRA under contract by the Italian Ministry of

Research [15]. The main objective of the program is to improve the National system and technology capabilities on liquid rocket engines for future space applications, with specific regard to LOX/LCH<sub>4</sub> technology. In this framework, the design and development of rocket engine demonstrators has been established, including intermediate test articles. A specific test article (referred to as Methane Thermal Properties, MTP, test article) has been designed to investigate the thermo-fluid dynamic behavior of methane inside a straight cooling channel with rectangular cross section, which can be representative of a regenerative system. In fact, the channel cross section has a dimension of 1 mm × 3 mm and the envisioned operative conditions are: maximum heat flux of the order of 10 MW/m<sup>2</sup>; inlet pressure ranging from 60 to 150 bar; inlet temperature of about 120 K; mass flow rate ranging from 10 to 30 g/s. A drawing of the experimental apparatus is shown in Fig. 1.

Goal of the present work is to investigate the transcritical methane flow inside the MTP test article of the HYPROB program. The flow inside the asymmetrically heated channel, characterized by rectangular cross section, is studied by a three-dimensional Conjugate Heat Transfer (CHT) model based on the numerical integration of the Navier–Stokes and Fourier's equations. Such a model is able to describe the whole cooling device composed by the coolant and the solid domain, whose heat input is represented by a condition of assigned entering heat flux. Computations with variable coolant pressure are performed in order to investigate the behavior of the MTP test article at different operative conditions. Moreover, also the effect of the wall surface roughness is investigated.

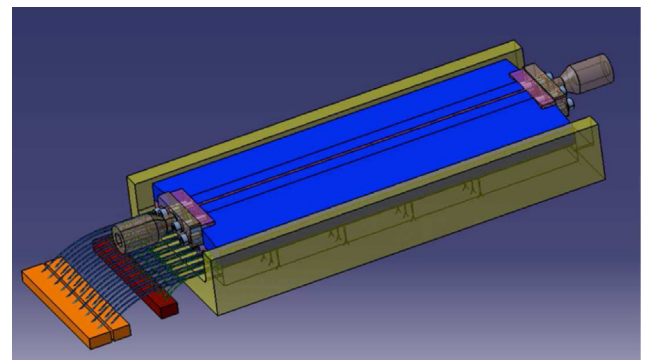


Fig. 1. Drawing of the MTP test article with electrical, mechanical and thermal insulation interfaces.

## 2. Test cases description

The geometric features of the MTP test article are: channel base  $b = 1$  mm, channel height  $h = 3$  mm, rib base  $t_w = 1$  mm, internal wall thickness  $s_w = 1$  mm and channel length  $L = 300$  mm, which corresponds to 200 hydraulic diameters. The imposed inlet conditions are the static temperature  $T_i$  and the mass flow rate  $\dot{m}$ , while the outflow condition is the static pressure  $p_e$ . The inner wall boundary condition is that of imposed heat flux  $q_w$  while adiabatic conditions are considered at the outer-wall interface for both the wall and the coolant and at the lateral walls. In particular, the imposed entering heat flux at the inner-wall is  $11.1 \text{ MW/m}^2$ . The schematic of the MTP test article with the enforced boundary conditions for the coolant and solid domain is shown in Fig. 2.

The coolant flow is supercritical methane while the solid material properties refer to a high-conductivity copper alloy with a thermal conductivity of  $k_w = 387.6 \text{ W/m K}$ , which is considered constant with temperature. Finally, wall surface is considered either smooth or with a certain level of equivalent-sand-grain roughness. To describe the different MTP operative conditions, different test cases have been considered, as described in Table 1. All the considered test cases share the same inlet temperature  $T_i = 120 \text{ K}$  and the same mass flow rate  $\dot{m} = 20 \text{ g/s}$ , while two different levels of exit pressure are considered,  $p_e = 100 \text{ bar}$  and  $p_e = 53.4 \text{ bar}$ , in order to describe the behavior of methane “sufficiently” far from the critical pressure, and in near-critical condition. Finally, different values of wall roughness, 0, 4.5, and  $10 \mu\text{m}$ , are considered.

## 3. Modeling

A Conjugate Heat Transfer (CHT) model based on the coupled numerical integration of the Navier–Stokes equations for the coolant flow and the Fourier’s equation for the heat conduction in the solid material is adopted in the present study [16,17]. In particular, the coolant-flow solver integrates the Reynolds Averaged Navier–Stokes equations, written in the conservation form, by a Godunov-type finite volume scheme, which is second-order accurate in space [18,19]. Its main feature is the capability to deal with any compressible fluid whose behavior is represented by a generic equation of state, as the fluid thermodynamic properties are stored in a look-up table. Turbulence is computed according to the one-equation model of Spalart–Allmaras for both smooth and rough

**Table 1**

Selected test cases.

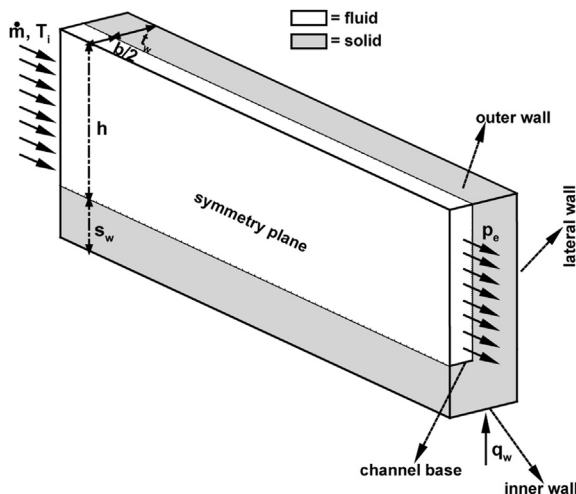
| TC# | $q_w$ [MW/m <sup>2</sup> ] | $p_e$ [bar] | $T_i$ [K] | $\varepsilon$ [ $\mu\text{m}$ ] | $\dot{m}$ [g/s] |
|-----|----------------------------|-------------|-----------|---------------------------------|-----------------|
| 1   | 11.1                       | 53.4        | 120       | 0                               | 20              |
| 2   | 11.1                       | 53.4        | 120       | 4.5                             | 20              |
| 3   | 11.1                       | 53.4        | 120       | 10                              | 20              |
| 4   | 11.1                       | 100         | 120       | 0                               | 20              |
| 5   | 11.1                       | 100         | 120       | 4.5                             | 20              |

surfaces [20,21]. Validation of the coolant-flow solver has been reported by comparison made with available experimental data on supercritical nitrogen [18,22] and transcritical hydrogen [10]. The coupled solution of the coolant flow and the wall heat conduction is achieved by an iterative procedure in which the wall temperature distribution obtained by a finite-volume solver, which is second-order accurate in space, of the Fourier’s thermal problem is used as boundary condition for the coolant flow computations [16]. Finally, the thermodynamic behavior of methane, which is the considered fluid in the present study, is described by the Modified–Benedict–Webb–Rubin equation of state and proper relations for the fluid viscosity and thermal conductivity, which are taken from Ref. [23].

## 4. Grid sensitivity

Numerical error assessment of the CHT simulations has been investigated by a grid sensitivity analysis performed by means of three levels of grid refinement. The most refined grid of the coolant domain, referred to as *fine grid*, is composed of  $64 \times 256 \times 64$  volumes along the half-base, the height and the axial direction of the channel. Taking advantage from the flow symmetry with respect to the vertical mid-plane (Fig. 2), only one-half of the physical domain is discretized. The volumes are clustered near the walls to accurately describe the turbulent boundary layers while the axial distribution is uniform. The computational fine grid of the solid domain is composed of  $4992 \times 64$  volumes along the cross section and the axial direction of the channel; moreover, in each coordinate direction, the mesh distribution is uniform. Also in this case only one-half of the physical domain is discretized (Fig. 2). The *medium grid* is obtained by removing one node out of two, in each coordinate direction and for both the coolant and the solid domain, from the fine grid. In the same way a *coarse grid* is obtained from the medium grid. Computational mesh of the channel cross section is presented in Fig. 3 for the medium and fine grids.

In particular, the grid sensitivity analysis has been performed on test cases 1 and 2 (Table 1) because of the relatively pronounced dependence of their solutions, especially the wall temperature, on the mesh refinements. The wall temperature distribution along the channel length and at the inner side of the cooling channel is shown in Fig. 4 for the three selected meshes. More specifically, the mentioned wall temperature is taken at the symmetry-plane, being negligible the temperature variations along the inner wall. Looking at Fig. 4, it is evident that the mesh refinement has a relevant impact on the wall temperature evaluation: for the test case with  $\varepsilon = 0 \mu\text{m}$ , wall temperature presents a peak value which increases by 95 K passing from the rough to the medium grid or by 24 K passing from the medium to the fine grid. For the test case with  $\varepsilon = 4.5 \mu\text{m}$ , these quantities reduces to 68 K and 14 K. Moreover, in both cases the position of the maximum value shifts upstream. The achieved asymptotic convergence behavior suggests that a further grid refinement would increase the peak values by  $\sim 6 \text{ K}$  in case of  $\varepsilon = 0 \mu\text{m}$  and by  $\sim 3 \text{ K}$  in case of  $\varepsilon = 4.5 \mu\text{m}$ . The relatively small improvement of the wall temperature prediction that could be achieved with a further grid refinement, allows one to adopt the fine mesh to describe such test cases.



**Fig. 2.** Schematic of the MTP test article and enforced boundary conditions.

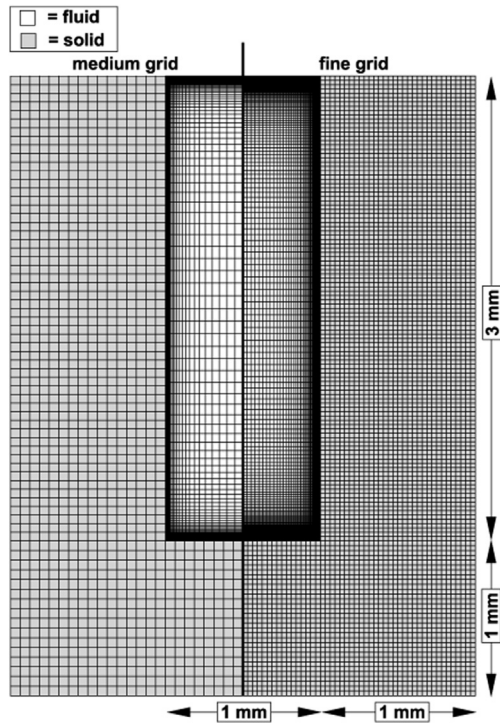


Fig. 3. Computational mesh schematic of the cooling-channel (fluid and solid domains).

The remaining test cases, which do not exhibit a peak value of the wall temperature, and for which only two grid levels have been considered, present a much milder dependence of the wall temperature on the grid refinement. In particular, the maximum wall temperature difference between coarse and medium grid is 8 K for test case 3, 30 K for test case 4, and 9 K for test case 5. The relatively small improvement of the wall temperature prediction that is expected with the fine grid, allows one to adopt the medium mesh to describe such test cases.

As for the other variables relevant to the analyzed problem (such as, coolant pressure drop and temperature gain), their numerical estimation presents a rather weak dependence on the mesh refinement. For instance, passing from the coarse to the medium grid, pressure drop variation is below 4% for the test cases

with smooth wall (TC1 and TC4), below 1.5% for the test cases with  $\varepsilon = 4.5 \mu\text{m}$  (TC2 and TC5), and below 0.1% for the test cases with  $\varepsilon = 10 \mu\text{m}$  (TC3). The reason for a less demanding mesh refinement with increasing wall roughness is that the wall roughness diminishes the gradients at wall and thus less numerical resolution is required. This can be seen also in Fig. 4, where wall temperature dependence on mesh refinement is more evident in case of  $\varepsilon = 0 \mu\text{m}$  rather than in case  $\varepsilon = 4.5 \mu\text{m}$ .

## 5. Results

### 5.1. Reference test case (TC1)

The thermal and fluid-dynamic behavior of TC1 is shown by temperature fields of both coolant and solid material in Fig. 5, and coolant specific heat at constant pressure fields in Fig. 6. In these figures, the evolution of the variables is shown at different cross-sections, from the inlet toward the exit, which are located every 6 cm along the streamwise direction. Methane enters the channel in a highly-compressed state; in fact, for the considered test case the resulting supercritical inlet pressure is  $p_i = 58.1$  bar and, due to the low-temperature inlet condition,  $T_i = 120$  K, its density is nearly  $415 \text{ kg/m}^3$ . Then, as it is heated-up by the entering heat flux from the hot walls located in the lower part of the channel section, the fluid accelerates while its pressure decreases due to the friction force acting on the channel wall and the entering heat flux. The pressure distribution (not shown here) is one-dimensional like, that is, it is only a function of abscissa  $x$ , regardless of the three-dimensional channel geometry and the asymmetric thermal boundary conditions. The one-dimensional behavior of pressure is a direct consequence of the constant cross-section, straight channel geometry. Fig. 5 shows that the high temperature flow region grows as the fluid moves from the inlet to the exit section, but it remains close to the hot wall along the whole length. As a consequence, the fluid far from the bottom wall is nearly at the inlet condition and an evident non-uniform thermal distribution occurs. Fig. 5 also permits to analyze the wall temperature distribution in the solid material. In every channel section, wall temperature exhibits a maximum value which is located on the inner wall and on the coolant symmetry-plane, that is, at the maximum distance from the solid ribs. This is because the thermal resistance of the coolant flow is larger than that of the solid ribs. For this reason the heat transfer rate, which entirely enters the coolant flow through its perimeter, flows toward the solid ribs more than toward the

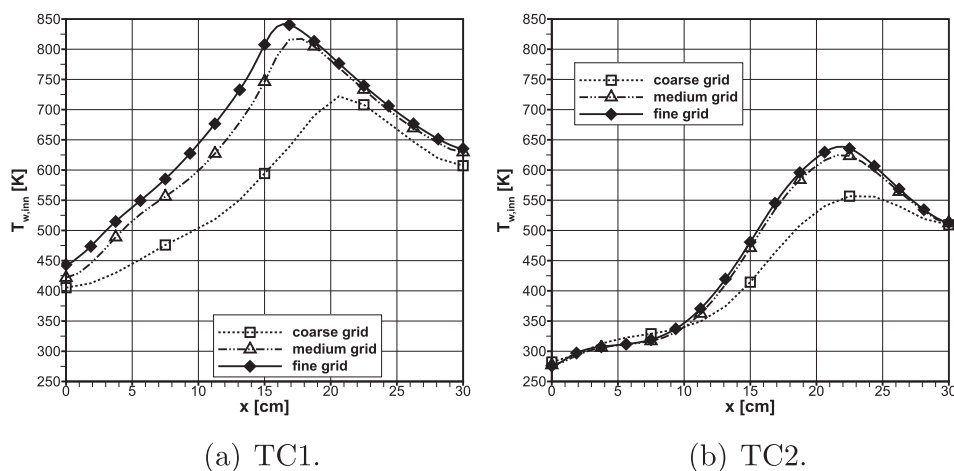


Fig. 4. Grid sensitivity on the inner-wall temperature.



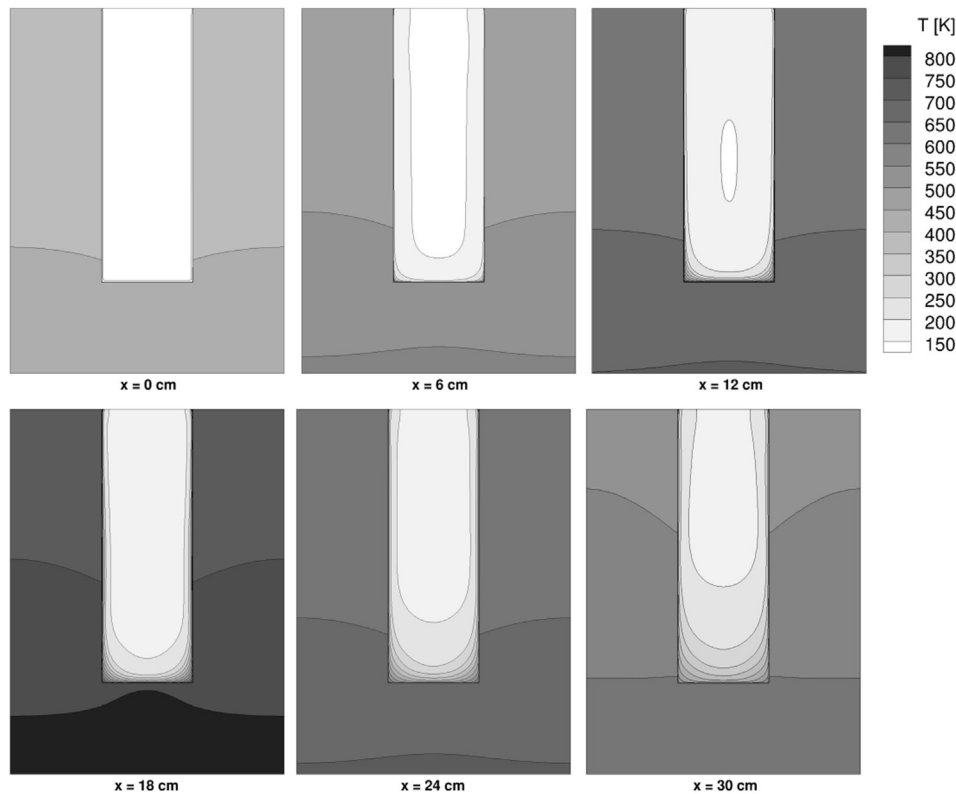


Fig. 5. Temperature fields at different channel sections.

channel base yielding a little temperature gradient in the transversal direction (i.e., base-direction); for instance, on the inner wall of the last channel section the wall temperature variation is nearly 3 K. Considering the wall temperature distribution along the channel length (see also Fig. 4(a)), a peak value of 842 K is found at section  $x = 16.4$  cm. This is a clear mark of the heat transfer deterioration that occurs for the present transcritical-fluid flow. It is interesting to note that the effect of the heat transfer deterioration is present on the whole cross section, not only at the inner wall. This is evidenced in the temperature field at  $x = 18$  cm, where solid temperature reaches a maximum everywhere in the channel section.

The specific heat at constant pressure along the channel length is presented in Fig. 6. Note that, because every channel section has a constant pressure while temperature crosses the pseudo-critical

value, ranging from the minimum value of 120 K to the maximum value of  $\approx 800$  K, thermodynamic parameters variation across a channel section is similar to the isobaric properties variation. Therefore, for a given cross section, constant pressure specific heat shows a maximum located at the pseudo-critical temperature. The presence of a maximum value of  $c_p$  is clearly seen in Fig. 6. Moreover, because the specific heat peak is more pronounced as the pressure approaches the critical value, the maximum value of  $c_p$  is present at the channel exit where the pressure reaches its minimum value ( $p_e = 53.4$  bar). At this pressure level the maximum value of specific heat is ten times higher than the perfect gas value (that is, the value at  $T \gg T_c$ ). Note that the relatively wide variation of the specific heat and the presence of a peak value, which virtually separates the “gas-like” region close to the walls to the “liquid-like” region, are the typical behavior of the transcritical-fluid flow. More

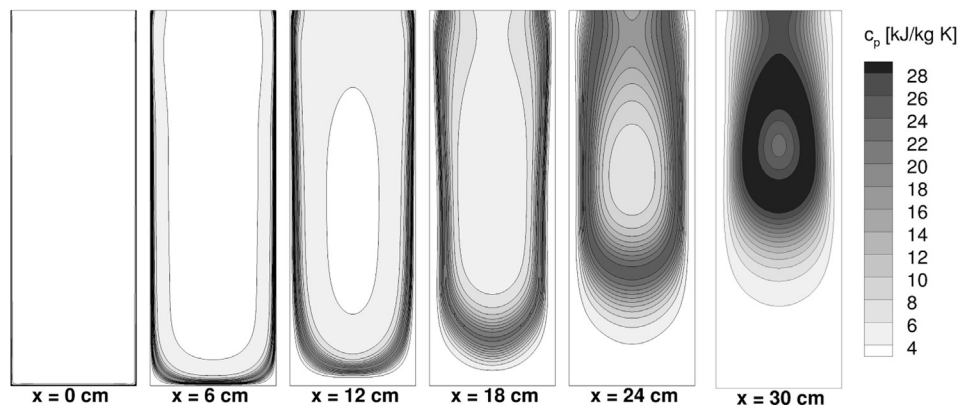


Fig. 6. Specific heat at constant pressure fields at different channel sections.

specifically, the heat transfer deterioration is generated by the gaseous film close to the wall which is characterized, because of its low density, by low thermal capacity. The gaseous film “insulates” the liquid in the core of the tube, causing a very high wall temperature and poor wall heat transfer. Downstream, the further fluid heating enlarges the low density region and, as a consequence, the bulk density decreases and the average fluid velocity increases. The increase of fluid velocity improves the heat transfer and, in turn, the wall temperature decreases. In case of TC1 the maximum wall temperature occurs at section  $x = 16.4$  cm.

### 5.2. Coolant pressure effect

The comparison of the results relevant to test cases 1 and 4 permits to appreciate the role of the fluid pressure level in the case of smooth wall. The comparison is of particular interest because of the relatively vicinity of the pressure levels to the critical pressure. In the case of TC1 the ratio of the exit pressure to the critical value is 1.16 and in the case TC4 it is 2.17. This suggests that the first test case will be much more affected by the large properties variation that occurs in the near-critical region than TC4. This is demonstrated in Fig. 7(a) where the inner-wall temperature distribution on the symmetry plane and along the channel length is presented. In case of TC1 the wall temperature presents a peak value of 842 K at the axial abscissa 16.4 cm while in the other case no peak value is present; in this case the wall temperature progressively increases along the axial abscissa and its maximum, located at the exit section, is 567 K. To better visualize the difference between the two considered test cases, the coolant bulk temperature and specific heat at constant pressure are presented in Fig. 7(b). The bulk temperature is defined as the mass-flux weighted average of the fluid temperature, over the tube cross section. In this figure, also the pseudo-critical temperature relevant to the two test cases is drawn. Note that the two values of the pseudo-critical temperature differ because of the different pressure level of the two test cases: it is nearly 196 K for TC1 and 218 K for TC4. Fluid bulk temperature exceeds this value at section  $x = 22.5$  cm for TC1 and  $x = 25.5$  cm for TC4. Here it is reminded that the fluid bulk temperature excess of the pseudo-critical value is the prerequisite for the heat transfer deterioration (Section 1). However, Fig. 7(a) clearly shows that deterioration takes place only in TC1. The different behavior between TC1 and TC4 must be sought in the pressure level of the two test cases. In case of TC1 the exit pressure is rather close to the

critical value (“only” 16% larger than the critical value) and thus the fluid properties variation is so intense to induce heat transfer deterioration; it is not the case of TC4. In fact, looking at Fig. 7(b), it is evident that the streamwise variation of the fluid properties is dramatic only for TC1: the specific heat reaches a value more than 7 times higher than at entrance in case of TC1 and less than 2 times in case of TC4. Looking at this figure, it is also important to notice that in case of TC1 the position of the specific heat peak ( $x = 22.5$  cm) is more downstream than the position of the maximum wall temperature ( $x = 16.4$  cm). This is because the performed analysis is based on fluid bulk variables, that are mass-flux weighted averages across the channel cross section; in reality, difference of the fluid properties is present also in the transversal directions, especially in the case of near critical fluids. This can be appreciated in Fig. 6, where it is shown that a maximum  $c_p$  contour line is present in every cross section the specific heat presents a peak value region. Finally, for what concerns the coolant pressure drop  $\Delta p_0$ , it is 2.8 bar in case of TC1 and 2.0 bar in the case of TC4. Thus, it is possible to say that, for a given coolant mass flow rate and entering heat transfer, the coolant pressure drop increases as the coolant pressure level approaches the critical value: pressure drop reduces by almost 30% passing from  $p_e/p_c = 1.16$  (TC1) to  $p_e/p_c = 2.17$  (TC4).

### 5.3. Roughness effect

The comparison between the results of TC1, TC2, and TC3 permits to appreciate, for a given pressure level and entering heat transfer, the role of the surface roughness level on the cooling performances. Surface is smooth in case of TC1 and is rough in cases TC2 ( $\varepsilon = 4.5 \mu\text{m}$ ) and TC3 ( $\varepsilon = 10 \mu\text{m}$ ). In Fig. 8(a) the inner-wall temperature distribution on the symmetry plane and along the channel length is presented for the three considered test cases. Test case with smooth wall (TC1) presents the peculiar behavior of the heat transfer deterioration which has been discussed in detail in the previous sections. Deterioration can also be recognized in TC2 because the wall temperature presents a peak value of 639 K at the axial abscissa 22.0 cm, but not in TC3 because the wall temperature progressively increases along the axial abscissa and its maximum, located at the exit section, is 396 K. The surface roughness effect on the wall temperature is easily found: thanks to the increased turbulence production with surface roughness, the wall temperature decreases. Moreover, there exists a threshold value of the roughness beyond which the heat transfer deterioration vanishes. For the

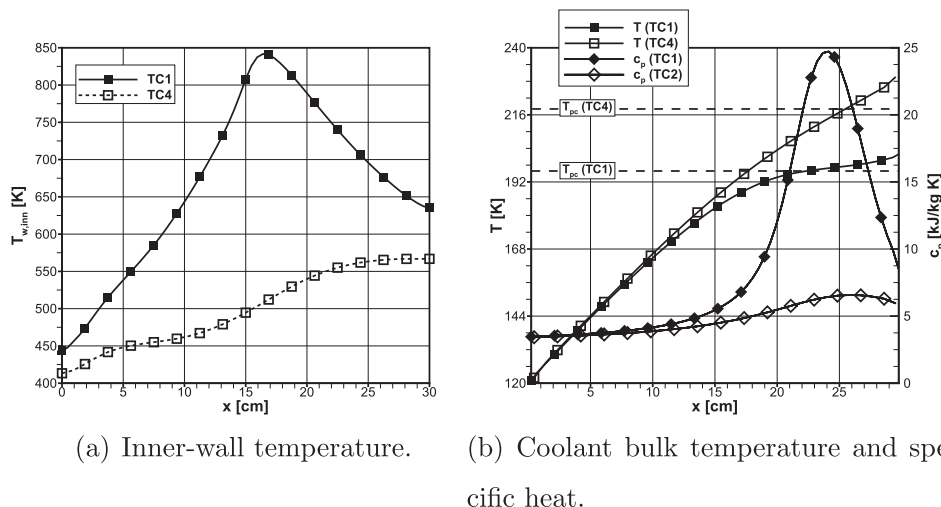


Fig. 7. Distribution along the channel length of the wall and coolant variables for TC1 and TC4.

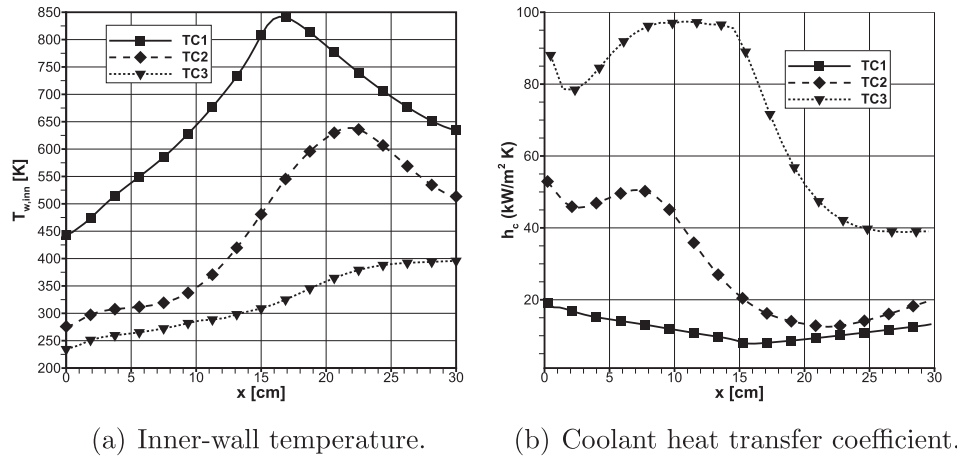


Fig. 8. Distribution along the channel length of the wall and coolant variables for TC1, TC2, and TC3.

considered test cases, the threshold is between  $\varepsilon = 4.5 \mu\text{m}$  and  $\varepsilon = 10 \mu\text{m}$ . The different wall temperature behaviors seen in Fig. 8(a) can be appreciated also with reference to Fig. 8(b), where the section-averaged heat transfer coefficient is shown for the three considered test cases. This variable is defined by:

$$h_c = \frac{\bar{q}_w}{\bar{T}_w - T} \quad (1)$$

where  $T$  is the fluid bulk temperature, and  $\bar{T}_w$  and  $\bar{q}_w$  are the average wall temperature and heat flux of a channel section. The  $h_c$  behavior in the very first part of the channel is driven by the mechanical boundary layer growth: initially it decreases and then it increases after reaching a minimum value. This is clearly visible in case of TC2 and TC3 (Fig. 8(a)), where the mentioned minimum value is reached at sections  $x = 2.3$  cm and  $x = 2.6$  cm. In case of TC1, the heat transfer coefficient minimum, located at 16.4 cm, is not related to the boundary layer growth but to the heat transfer deterioration. In fact, the minimum value of  $h_c$  corresponds to the maximum value of the wall temperature (Fig. 8(a)) and is equal to  $7.8 \text{ kW/m}^2 \text{ K}$ , that is, almost three times lower than the inlet value and two times lower than the exit value. In the other two test cases, namely TC2 and TC3, the heat transfer coefficient presents a substantially different behavior with respect to that of TC1. In particular, after the boundary layer growth, the heat transfer coefficient increases up to a maximum and then decreases to a minimum value. More specifically, in case of TC2 the maximum value is equal to  $50.5 \text{ kW/m}^2 \text{ K}$ , located at  $x = 7.3$  cm, and the minimum is  $12.5 \text{ kW/m}^2 \text{ K}$ , located at  $x = 22.0$  cm. Note that this minimum  $h_c$  value, being larger than that of TC1, causes a maximum wall temperature which is about 200 K less than the maximum wall temperature of TC1. This is because of the beneficial effect of the surface roughness on the wall temperature distribution (Fig. 8(a)). For the same reason, TC3 presents an even larger heat transfer coefficient characterized by a plateau of  $\sim 97 \text{ kW/m}^2 \text{ K}$ , between sections at  $x \approx 8$  cm and  $x \approx 14$  cm, followed by a monotonic decrement up to the exit value of  $\sim 39 \text{ kW/m}^2 \text{ K}$ . The larger large value of  $h_c$  in case of TC3, induced by the relatively high value of surface roughness, prevents from heat transfer deterioration (Fig. 8(a)).

Of course, the wall temperature reduction, up to the suppression of the heat transfer deterioration, which can be realized thanks to the increment of the surface roughness level, is paid in terms of pressure drop  $\Delta p_0$ . In fact, while it is 2.8 bar in case of  $\varepsilon = 0 \mu\text{m}$

(TC1), it raises up to 3.5 bar (25%) in case of  $\varepsilon = 4.5 \mu\text{m}$  (TC2) and up to 5.2 bar (86%) in case of  $\varepsilon = 10 \mu\text{m}$  (TC3).

Roughness effect has been investigated also in case of higher pressure, that is,  $p_e = 100$  bar. In particular, surface is smooth in case of TC4 and has a roughness level of  $\varepsilon = 4.5 \mu\text{m}$  in the case TC5. As already seen in Section 5.2, the wall temperature distribution of TC4 (Fig. 7(a)) is not affected by the heat transfer deterioration and the maximum wall temperature is 567 K (at the exit section). The surface roughness level of  $\varepsilon = 4.5 \mu\text{m}$  is able to reduce the wall temperature of  $\sim 140$  K all along the channel length; the resulting maximum value, always at the exit, is 431 K. As for the pressure drop, it is 2.0 bar in case of  $\varepsilon = 0 \mu\text{m}$  (TC4) and it raises up to 3.2 bar (60%) in case of  $\varepsilon = 4.5 \mu\text{m}$  (TC5). As already mentioned in Section 5.2, for a given coolant mass flow rate and entering heat transfer, the coolant pressure drop increases as the coolant pressure level approaches the critical value. Here it is also demonstrated that the roughness effect on the pressure drop is more marked in case of larger coolant pressure level: in fact, passing from  $\varepsilon = 0 \mu\text{m}$  to  $\varepsilon = 4.5 \mu\text{m}$ ,  $\Delta p_0$  increases by 60% in case of  $p_e = 100$  bar and by 25% in case of  $p_e = 53.4$  bar.

## 6. Conclusion

The present study has been focused on the near-critical operative condition of methane inside a rectangular cooling channel, which can be representative of a rocket engine regenerative system. For the given nominal heat input, mass flow rate and inlet subcritical temperature, deterioration has been encountered in case of smooth surfaces and exit-to-critical pressure ratio equal to 1.16. In this case the maximum wall temperature is nearly 850 K. Different coolant pressure and surface roughness levels have been considered in order to understand their influence on the heat transfer capability of the cooling system. Numerical results have highlighted that the heat transfer deterioration is mitigated either in case of higher coolant pressure or higher surface roughness. In the latter case, an important penalty in terms of coolant pressure drop occurs. On the other hand, the increase to an exit-to-critical pressure ratio equal to 2.17 leads to a maximum wall temperature of almost 570 K and to a 30% lower pressure drop. Note that heat transfer deterioration does not occur at a surface roughness level of  $\varepsilon = 10 \mu\text{m}$ , which however yields to an increase of almost 90% of the coolant pressure drop. At lower roughness level, deterioration does not disappears but its effect on the maximum wall temperature is significantly reduced.

## Acknowledgements

This work has been carried out within the HYPROB program, funded by the Italian Ministry of University and Research (MIUR n. 1090 22/10/2008) whose financial support is appreciated.

## References

- [1] R.J. Quentmeyer, Experimental fatigue life investigation of cylindrical thrust chambers, AIAA Paper 1977-893, in: 13th AIAA/SAE Propulsion Conference, Jul. 1977.
- [2] K. Xu, B. Ruan, H. Meng, A thermal performance factor for evaluation of active engine cooling with asymmetric heating, *Appl. Therm. Eng.* 73 (1) (2014) 349–354.
- [3] G.P. Sutton, History of Liquid Propellant Rocket Engines, AIAA, United States, 2006.
- [4] P. Caisso, A. Souchier, C. Rothmund, P. Alliot, C. Bonhomme, W. Zinner, R. Parsley, T. Neill, S. Forde, R. Starke, W. Wang, M. Takahashi, M. Atsumi, D. Valentian, A liquid propulsion panorama, *Acta Astronaut.* 65 (11–12) (2009) 1723–1737.
- [5] V. Kurganov, Y. Zeigarnik, I. Maslakova, Heat transfer and hydraulic resistance of supercritical-pressure coolants, part i: Specifics of thermophysical properties of supercritical pressure fluids and turbulent heat transfer under heating conditions in round tubes (state of the art), *Int. J. Heat Mass Transf.* 55 (11–12) (2012) 3061–3075.
- [6] J. Jackson, Fluid flow and convective heat transfer to fluids at supercritical pressure, *Nucl. Eng. Des.* 264 (2013) 24–40.
- [7] R. Cook, Methane Heat Transfer Investigation, NASA, 1984. CR-171051.
- [8] H. Kawashima, H. Negishi, T. Tomita, K. Obase, T. Kaneko, Verification of prediction methods for methane heat transfer characteristics, AIAA Paper 2012-4120, in: 48th AIAA/ASME/SAE/ASEE Joint Propulsion Conference, 2012.
- [9] H. Gu, H. Li, H. Wang, Y. Luo, Experimental investigation on convective heat transfer from a horizontal miniature tube to methane at supercritical pressures, *Appl. Therm. Eng.* 58 (1–2) (2013) 490–498.
- [10] M. Pizzarelli, A. Urbano, F. Nasuti, Numerical analysis of deterioration in heat transfer to near-critical rocket propellants, *Numer. Heat Transf. Part A Appl.* 57 (5) (2010) 297–314.
- [11] Y.-Z. Wang, Y.-X. Hua, H. Meng, Numerical studies of supercritical turbulent convective heat transfer of cryogenic-propellant methane, *J. Thermophys. Heat Transf.* 24 (3) (2010) 490–500.
- [12] M. Pizzarelli, F. Nasuti, M. Onofri, Cfd analysis of transcritical methane in rocket engines cooling channels, *J. Supercrit. Fluids* 62 (2012) 79–87.
- [13] H. Negishi, Y. Daimon, H. Kawashima, N. Yamanishi, Flowfield and heat transfer characteristics of cooling channel flows in a methane-cooled thrust chamber, AIAA Paper 2012-4122, in: 48th AIAA/ASME/SAE/ASEE Joint Propulsion Conference, 2012.
- [14] L. Wang, Z. Chen, H. Meng, Numerical study of conjugate heat transfer of cryogenic methane in rectangular engine cooling channels at supercritical pressures, *Appl. Therm. Eng.* 54 (1) (2013) 237–246.
- [15] R. Votta, F. Battista, M. Ferraiuolo, P. Roncioni, P. Vernillo, V. Salvatore, P. De Matteis, Design of an experimental campaign on methane regenerative liquid rocket engine cooling system, AIAA Paper 2013-4146, in: 49th AIAA/ASME/SAE/ASEE Joint Propulsion Conference, Jul. 2013.
- [16] M. Pizzarelli, F. Nasuti, M. Onofri, Coupled wall heat conduction and coolant flow analysis for liquid rocket engines, *J. Propuls. Power* 29 (1) (2013) 34–41.
- [17] M. Pizzarelli, F. Nasuti, M. Onofri, Effect of cooling channel aspect ratio on rocket thermal behavior, *J. Thermophys. Heat Transf.* 28 (3) (2014) 410–416.
- [18] M. Pizzarelli, F. Nasuti, R. Paciorri, M. Onofri, Numerical analysis of three-dimensional flow of supercritical fluid in asymmetrically heated channels, *AIAA J.* 47 (11) (2009) 2534–2543.
- [19] M. Pizzarelli, Effectiveness of spalart-allmaras turbulence model in analysis of curved cooling channels, *AIAA J.* 51 (9) (2013) 2158–2167.
- [20] P. Spalart, S. Allmaras, A one-equation turbulence model for aerodynamic flows, *La Rech. Aerosp.* (1) (1994) 5–21.
- [21] B. Aupoix, P.R. Spalart, Extensions of the spalart-allmaras turbulence model to account for wall roughness, *Int. J. Heat Fluid Flow* 24 (4) (2003) 454–462.
- [22] M. Pizzarelli, F. Nasuti, M. Onofri, Analysis of curved cooling channel flow and heat transfer in rocket engines, *J. Propuls. Power* 27 (5) (2011) 1045–1053.
- [23] B. Younglove, J. Ely, Thermophysical properties of fluids. II. Methane, ethane, propane, isobutane, and normal butane, *J. Phys. Chem. Ref. Data* 16 (4) (1987) 577–798.